STUDY + AI · GRADES 7–12 · FREE CARD DECK

Study Moves

Ten ways to study that research shows work. Pick what you want to do, then use the move with a book, pen and paper. Your teacher can prepare it with AI, and if you are 13 or older you can study with AI.

> **Not sure? Start here: Test yourself.** Close the book and pull what you have learned out of your memory, instead of reading it again.

What do you want to do?

Remember it
1. Test yourself, 2. Spread it out

Tell similar things apart
3. Mix it up

Understand it
4. Explain it simply, 5. Ask why, 6. Draw it, 7. Find an example

Solve problems
8. Study a solved example, 3. Mix it up

Fix your mistakes
9. Fix your mistakes

Start a new topic
10. Guess first

Why good studying feels harder

> Good study moves feel worse while you use them. Testing yourself is slower than reading again, and you get things wrong. Reading again feels smooth, so it feels like learning. Most students re-read rather than test themselves (Karpicke, Butler & Roediger 2009), and in one study physics students learned more in lessons that made them work hard but felt they had learned less (Deslauriers et al. 2019). If a move feels hard, that is usually a sign it is working.

The ten moves

1. Test yourself **Strong**
2. Spread it out **Strong**
3. Mix it up **Promising**
4. Explain it simply **Promising**
5. Ask why **Promising**
6. Draw it **Promising**
7. Find an example **Promising**
8. Study a solved example **Strong**
9. Fix your mistakes **Strong**
10. Guess first **Promising**

How strong is the evidence?

Strong Meta-analyses, including studies in real classrooms, show that studying this way improves learning.

Promising Controlled studies show gains, but there are fewer of them, results vary, or they depend on the material.

Framework A sensible method built on related research, with no direct studies of the method itself yet.

MOVE 1 · GOOD FOR: REMEMBER IT

Test yourself

Strong Retrieval practice

Close the book and pull what you have learned out of your memory, instead of reading it again.

HOW TO DO IT

1. Close the book and put your notes away.
2. Write down everything you remember, or answer questions: flashcards, the questions at the end of the chapter or an old test.
3. Open the book and check. Mark what you missed or got wrong.
4. Next time, start with what you missed.

> **Example: History** · Grade 8
>
> Before a test on the French Revolution, close your notes and write down the main causes, three key dates and three key people from memory. Check against your notes. Whatever you missed is where you start tomorrow.

HOW IT FEELS

> It is slower than reading again, and you will get things wrong. That is the point: every time you pull something out of memory, it gets easier to find the next time.

WITH AI: YOUR TEACHER

Your teacher pastes in the topic and gets ten questions with answers, to print for you to answer with the book closed.

For the AI (teacher). Ten short questions that make students pull the topic's key content out of memory: mostly short answers, some 'explain why' questions, easiest first. Students answer on paper with the book closed, then check against the answers.

WITH AI: YOU (13+)

The AI quizzes you one question at a time and only gives the answer after you have tried.

For the AI (student). Quiz me on the topic with short questions, one at a time: mostly facts, some 'explain why'. After each answer, tell me whether I was right and give the correct answer in one or two sentences. After ten questions, list the ones I got wrong so I can study them.

WATCH OUT

Check your answers every time, so that a wrong answer does not stick. Testing yourself helps most with facts. To understand how something works, add Explain it simply or Ask why.

Evidence. Testing yourself is one of the two best-supported study techniques (Dunlosky et al. 2013). A meta-analysis of classroom studies confirms that it improves learning in school (Yang et al. 2021). In real classrooms the effect is smaller than in lab studies, but still clear (Ruitenburg et al. 2026).

MOVE 2 · GOOD FOR: REMEMBER IT

Spread it out

Strong Spaced practice

Study in short sessions on different days, instead of one long session the night before.

HOW TO DO IT

1. Split your study time into short sessions of 15 to 30 minutes on different days.
2. Start each session by testing yourself on what you studied last time.
3. Leave a day or more between sessions, and keep coming back even after it feels easy.
4. For a test, start at least a week before. Use the last evening for a quick check, not for reading everything for the first time.

> **Example: Modern languages** · Grade 9
>
> Instead of learning 40 Spanish words on Sunday evening, learn 10 on Monday. On Wednesday, test yourself on Monday's words before you learn 10 more, and so on. On Sunday, test yourself on all 40.

HOW IT FEELS

> Each time you come back, you will have forgotten some of it, and that feels bad. Pulling it back is exactly what makes it last.

WITH AI: YOUR TEACHER

Your teacher gets a two-week plan of four short sessions, each starting with questions on what came before.

For the AI (teacher). A two-week plan of four short study sessions of 10 to 15 minutes, labelled Day 1, Day 3, Day 7 and Day 14. Each session has five questions: two on content from earlier sessions and three on this session's part of the topic.

WITH AI: YOU (13+)

The AI helps you plan short sessions before your test, and starts each session with questions on earlier parts.

For the AI (student). First ask me when my test is and how many minutes a day I can study, then suggest a plan with short sessions on different days. Then quiz me on the first part of the topic. When I tell you I am back for a later session, start with questions on the earlier parts.

WATCH OUT

There is no perfect gap: the further away the test, the longer the gaps can be. Spreading it out works best together with testing yourself.

Evidence. Spreading practice out is the other of the two best-supported study techniques (Dunlosky et al. 2013). A meta-analysis finds that it clearly beats doing the same practice in one go (Latimier, Peyre & Ramus 2021). In real classrooms the effect is smaller, but still positive (Ruitenburg et al. 2026).

Mix it up

Promising Interleaving

Mix different types of problems or topics in one session, so that you have to choose the method yourself.

HOW TO DO IT

1. Learn each type on its own first.
2. Then practise with the types mixed in a random order, not all of one type in a row.
3. Before you solve each one, say which type it is and how you can tell.
4. Check your answers and note which types you mixed up.

> **Example: Maths** · Grade 9
>
> Instead of 20 area problems followed by 20 volume problems, mix them: the area of a circle, the volume of a cylinder, the area of a triangle, the volume of a box. Before each one, write 'area' or 'volume' and the formula you need.

HOW IT FEELS

> Mixing feels messier, and you will be slower and make more mistakes while you practise. That is because you have to decide which method to use, which is exactly what a test asks of you.

WITH AI: YOUR TEACHER

Your teacher gets twelve mixed practice questions where you first name the type and then solve it.

For the AI (teacher). Twelve practice questions that mix three types from this topic that students often confuse, in shuffled order. For each one, students first write which type it is and how they can tell, then answer it.

WITH AI: YOU (13+)

The AI gives you mixed questions one at a time and asks which type each one is before you solve it.

For the AI (student). Give me practice questions that mix different types from this topic in a random order, one at a time. For each one, first ask me which type it is and how I can tell. Then let me solve it, and tell me whether I was right.

WATCH OUT

Only mix things you have already learned once. It helps most when the types look alike and are easy to confuse. For learning new words, practising them in blocks may work better.

Evidence. A meta-analysis finds that mixing helps on average, most when learning to tell similar things apart, such as painting styles, less in maths, and not for learning words (Brunmair & Richter 2019). It helped in real classrooms too, but that finding rests on only ten studies (Ruitenburg et al. 2026).

MOVE 4 · GOOD FOR: UNDERSTAND IT

Explain it simply

Promising · The Feynman technique

Explain the topic in simple words, as if to someone younger, and find the parts you cannot explain yet.

HOW TO DO IT

1. Decide before you study that you will explain it afterwards.
2. Close the book. Explain the topic out loud or on paper as if to a ten-year-old, without difficult words.
3. Mark where you got stuck or used a word you could not explain. Go back to the book for just those parts.
4. Explain it again, shorter and clearer.

> **Example: Biology** · Grade 8
>
> Explain photosynthesis to a ten-year-old. If you say 'chlorophyll' and cannot say what it does, that is the part to look up before you try again.

HOW IT FEELS

> You will find gaps you did not know you had, and that is uncomfortable. Finding them now is much better than finding them in the test.

WITH AI: YOUR TEACHER

Your teacher gets an explain-it task, three 'but why?' questions a beginner would ask, and a checklist for checking your explanation.

For the AI (teacher). A task where students explain the topic to a ten-year-old in at most 150 words with the book closed, three 'but why?' questions a curious beginner would ask, and a five-point checklist students use to check their own explanation or a classmate's.

WITH AI: YOU (13+)

The AI plays a curious beginner who only asks questions. It never explains the topic itself.

For the AI (student). Play a curious ten-year-old who knows nothing about the topic. Ask me to explain it. Then ask 'but why?' or 'what does that word mean?' about the parts I explain vaguely. Never explain the topic yourself. When I say I am done, tell me which part of my explanation was least clear.

WATCH OUT

Decide before you start studying that you will explain it: that is when the effect is largest. Check your explanation against the book, because a confident explanation can still be wrong.

Evidence. There are no studies of the Feynman technique by name. It combines two things that are well studied: explaining to yourself while you learn, which helps across many studies (Bisra et al. 2018), and learning in order to teach someone, which helps most when you know before you study that you will teach (Kobayashi 2024).

MOVE 5 · GOOD FOR: UNDERSTAND IT

Ask why

Promising Elaborative interrogation

Ask yourself why each fact is true, and answer with what you already know.

HOW TO DO IT

1. Take one fact you are learning.
2. Ask: why is this true? Why does it happen?
3. Answer in your own words, using what you already know.
4. Check your answer against the book or your notes.

> **Example: Geography** · Grade 7
>
> Fact: it rains much more on the west coast of Norway than east of the mountains. Why? Wet air from the sea is pushed up the mountains, cools and drops its water as rain before it reaches the other side.

HOW IT FEELS

> It is slower than reading, and sometimes you will not know the answer. Trying to answer still helps you understand and remember.

WITH AI: YOUR TEACHER

Your teacher gets eight key facts, each with a 'why?' question and a short model answer to check against.

For the AI (teacher). Eight key facts from the topic, each followed by a 'Why is this true?' or 'Why does this happen?' question that students answer from what they already know. In ANSWERS, give a short model explanation for each.

WITH AI: YOU (13+)

The AI gives you one fact at a time, asks you why it is true and then tells you what was missing in your reasoning.

For the AI (student). Give me one fact from the topic at a time and ask me why it is true or why it happens. Let me answer from what I know. Then tell me what was right and what was missing in my reasoning, in two or three sentences.

WATCH OUT

It works best when you already know something about the topic. Always check your reasons: a wrong explanation can stick just as well as a right one.

Evidence. A large review rates asking why as moderately useful (Dunlosky et al. 2013). A meta-analysis finds a medium average effect, mostly on tests of facts given soon after studying (Donoghue & Hattie 2021).

MOVE 6 · GOOD FOR: UNDERSTAND IT

Draw it

Promising | Dual coding

Turn what you read into a drawing, such as a diagram, a timeline or a flowchart, and explain it in words.

HOW TO DO IT

1. Read a section, then close the book.
2. Draw it: a labelled diagram, a timeline, a flowchart or a map of how the ideas connect.
3. Explain your drawing in words, out loud or in two sentences.
4. Compare with the book and fix what is missing or wrong.

> **Example: Physics** · Grade 9
>
> After reading about electric circuits, close the book and draw a battery, a switch and two lamps in series. Mark the direction of the current and explain what happens to the other lamp when one breaks.

HOW IT FEELS

> Drawing takes longer than highlighting, and your first drawing will be messy. You are not making art: you are working out how the parts connect.

WITH AI: YOUR TEACHER

Your teacher gets a drawing task: which kind of diagram fits the topic, what it must show, and a model to compare with.

For the AI (teacher). A drawing task: which kind of drawing fits this topic best (labelled diagram, timeline, flowchart or concept map) and what it must include. Students draw it from memory and explain it in two sentences. In ANSWERS, describe a model drawing in words and list what a complete drawing shows.

WITH AI: YOU (13+)

The AI tells you what to draw, asks you to describe your drawing and points out what is missing, without drawing it for you.

For the AI (student). Suggest which kind of drawing fits the topic (diagram, timeline, flowchart or concept map) and ask me to draw it on paper from memory. Then ask me to describe my drawing in words. Tell me what is missing or wrongly connected, but do not draw it or give me the full answer.

WATCH OUT

The drawing must show how things connect. Decoration and colours do not help on their own. Check your drawing against the book.

Evidence. Pairing words with a clear picture helps learning in many studies (Cromley & Chen 2025). Making your own drawing while you learn also helps on average (Fiorella & Zhang 2018), but the results vary with how the drawing is done.

MOVE 7 · GOOD FOR: UNDERSTAND IT

Find an example

Promising · Concrete examples

Find your own examples of an abstract idea, and work out what they have in common.

HOW TO DO IT

1. Take an abstract idea, such as 'opportunity cost' or 'irony'.
2. Find two or three examples from different places: the book, your own life, the news.
3. Say what the examples have in common. That is the idea.
4. Find one thing that looks similar but is not an example, and say why.

> **Example: Economics** · Grade 12
>
> Opportunity cost: going to football practice means missing a friend's party; a town that builds a swimming pool cannot spend that money on a library. What they share: choosing one thing means giving up the next best thing.

HOW IT FEELS

> Thinking of your own examples is harder than reading the ones in the book. That effort is what makes the idea yours.

WITH AI: YOUR TEACHER

Your teacher gets examples of two key ideas, a task to find your own, and a tricky case that is not an example.

For the AI (teacher). Two abstract ideas from the topic. For each: one concrete example explained in two sentences, a task asking students to find two examples of their own from different areas of life, and one case that looks similar but does not fit, with the reason in ANSWERS.

WITH AI: YOU (13+)

The AI names an idea, asks for your own examples and tells you whether they really fit.

For the AI (student). Name one idea from the topic and ask me for two examples of my own from different areas of life. Tell me whether each one really fits and why. Then give me a tricky case that looks like an example but is not, and ask me why it does not fit.

WATCH OUT

Use more than one example and make them different, so that you do not mistake a surface detail for the idea. Check your examples with the book or your teacher.

Evidence. Studying several concrete examples helps students understand and use abstract ideas in controlled studies (Rawson, Thomas & Jacoby 2015). There is no meta-analysis yet, and most of the studies are with university students.

MOVE 8 · GOOD FOR: SOLVE PROBLEMS

Study a solved example

Strong | Worked examples

Study a fully solved example step by step before you solve similar problems on your own.

HOW TO DO IT

1. Find a fully solved example: in the book, from your teacher or in your notes.
2. Go through it step by step and say why each step is done.
3. Cover the solution and solve the same problem yourself.
4. Then solve a similar problem on your own. Move on to problems without examples as you get better.

> **Example: Chemistry** · Grade 10
>
> Study a solved example of balancing $H_2 + O_2 \rightarrow H_2O$ step by step. Cover it and balance it yourself. Then try $CH_4 + O_2 \rightarrow CO_2 + H_2O$ on your own.

HOW IT FEELS

> Looking at the solution can feel like cheating. But when a topic is new, studying a solution teaches you more than being stuck on your own.

WITH AI: YOUR TEACHER

Your teacher gets solved examples with every step explained, each followed by a similar problem for you to solve.

For the AI (teacher). Two pairs of problems. Each pair is a fully worked example with every step explained in one line, followed by a similar problem for students to solve on their own. Then two problems with no example. Give full solutions in ANSWERS.

WITH AI: YOU (13+)

The AI shows one solved example, gives you a similar problem and points to your first wrong step without fixing it for you.

For the AI (student). Show me one fully worked example from this topic, step by step, with one line on why each step is done. Then give me a similar problem to solve on my own. Check my solution and point to my first wrong step without fixing it for me. When I get one right, give me a new problem without an example.

WATCH OUT

This helps most when a topic is new. As you get better, solving problems on your own helps more than studying examples. Most of the research is in maths.

Evidence. A meta-analysis of studies in maths, from primary school to university, finds that learning from worked examples beats solving problems alone (Barbieri et al. 2023). The benefit is largest for beginners and shrinks as you gain skill (Sweller, van Merriënboer & Paas 2019).

MOVE 9 · GOOD FOR: FIX YOUR MISTAKES

Fix your mistakes

Strong Feedback

Check every answer, find out why the wrong ones were wrong, and test yourself on them again later.

HOW TO DO IT

1. After a test or quiz, check every answer against the key or the book as soon as you can.
2. For each mistake, write the right answer and why yours was wrong.
3. A few days later, test yourself again on just the questions you got wrong.

> **Example: English** · Grade 11
>
> Your essay comes back with the comment 'quotes without analysis'. Rewrite one paragraph so that each quote is followed by what it shows and why it matters. Then check the paragraph against the comment.

HOW IT FEELS

> Looking at your mistakes is uncomfortable, and it is tempting to look only at the score. But your mistakes show you exactly what to study next.

WITH AI: YOUR TEACHER

Your teacher gets a quiz with an answer key that explains each answer, a correction sheet and three retest questions for later.

For the AI (teacher). A ten-question quiz on the topic, an answer key that explains each answer in one sentence, a short correction sheet where students write the right answer and why theirs was wrong for each mistake, and three retest questions on the same content, worded differently, for a few days later.

WITH AI: YOU (13+)

The AI quizzes you, asks you to find your own mistakes before it explains them, and ends with new questions on what you got wrong.

For the AI (student). Quiz me with ten short questions, one at a time, without telling me yet whether I was right. After all ten, go through my wrong answers one by one: first ask me to find the mistake myself, then explain it briefly. End with three new questions on what I got wrong.

WATCH OUT

Check soon, and look at more than the score. A mistake you only notice, without writing down the right answer, is easy to make again.

Evidence. Feedback is one of the best-supported ways to improve learning, especially when it tells you what to do next (Wisniewski, Zierer & Hattie 2020). Testing yourself also works better when you check the right answer afterwards (Rowland 2014).

MOVE 10 · GOOD FOR: START A NEW TOPIC

Guess first

Promising Pretesting

Before a new topic, try to answer questions about it, even if you have to guess.

HOW TO DO IT

1. Before a new chapter or lesson, find questions on it: the questions at the end of the chapter, or ones your teacher gives you.
2. Answer them all, even if you have to guess. Do not look anything up.
3. Then read the chapter or follow the lesson, and look for the answers.
4. Check your guesses afterwards.

> **Example: Science** · Grade 7
>
> Before the chapter on the solar system, guess: which planet is the hottest, and why do we have seasons? Then read and check. It is Venus, not Mercury, and seasons come from the tilt of the Earth, not its distance from the Sun.

HOW IT FEELS

> Guessing wrong feels silly. Wrong guesses still help: they show your brain what to look for when you learn it.

WITH AI: YOUR TEACHER

Your teacher gets six questions to try before the topic is taught, aimed at its key ideas.

For the AI (teacher). Six questions to try before the topic is taught, mixing multiple choice and short answers, aimed at the key ideas students will meet. Use common wrong ideas as the wrong options. Students answer even if they have to guess, and keep their answers to check after the lesson.

WITH AI: YOU (13+)

The AI asks you questions on a topic you have not learned yet and keeps the answers until after you have studied.

For the AI (student). I have not learned this topic yet. Ask me five questions about its key ideas, one at a time, and tell me that guessing is fine. Do not tell me the answers yet. At the end, list the five questions so I can look for the answers when I study.

WATCH OUT

It mostly helps you learn what the questions were about. Answering, even with a guess, helps more than just reading the questions.

Evidence. A meta-analysis finds that trying questions before learning helps you learn the answers to those questions, more when you answer than when you only read them, with a smaller benefit for other content (St. Hilaire et al.).

Your own prompt: teacher

Fill in the blanks, then type the prompt into any AI tool: Copilot, Gemini, ChatGPT or Claude. Leave out the line "Students already know" if you have nothing to add. Check the material before you hand it out: the AI proposes, you decide.

```
Do not include student names or student work in this conversation.

You are helping a Grade _____ _____ teacher prepare study material
that students will use on their own, on paper and without AI.

Study move: _____ (the move's name and one-line description from
its card)
What to make: _____ (the "For the AI (teacher)" text from
its card)

Topic: _____
Students already know: _____

Use only this topic's content. Do not repeat a question. If you write multiple-choice
questions, make all options about the same length.

Reply in plain text, no tables, with exactly these four parts:
MOVE: the move's name and one sentence, written to students, on why it works.
FOR STUDENTS: the material, written to the students and ready to print.
ANSWERS: the answers or a checklist students use to check their own work, numbered Q1:,
Q2: and so on to match.
CHECK: every fact, figure, date or source in FOR STUDENTS or ANSWERS that the teacher
should verify, one per line. Write "None" if there are none.
```

Faster online: the prompt builder at choosewise.education/study-moves fills this in for you. Students use the material on paper, without AI.

Your own prompt: student

Fill in the blanks, then type the prompt into your AI tool. Leave out the line "What I already know" if you have nothing to add. The AI asks and you answer: you do the thinking.

```
Do not include your name or other personal details in this conversation.

You are my study partner. I am a Grade _____ student studying _____.
Topic: _____
What I already know: _____

Study move: _____ (the move's name and one-line description from
its card)
Your role: _____ (the "For the AI (student)" text from its
card)

Rules for the whole conversation:
- Give me one question or task at a time, then wait for my answer.
- Never give me the answer to a question before I have tried it. If I am stuck, give a
small hint first.
- Keep every message short.
- Do not write my homework, essays or answers for me, even if I ask.
- If you are not sure a fact is right, say so. I will check facts in my textbook or
notes.

Start now.
```

AI tools and age rules (November 2026). **Copilot Chat:** Students aged 13 and over, through the school's own Microsoft 365 account. The school's administrator must set each student's age group and consent. **Gemini for Education:** Open to education users of all ages, with a stricter experience for under-18s. The school's administrator must turn it on. **ChatGPT:** Ages 13 and over on personal accounts, with a parent's or guardian's permission under 18. ChatGPT for Teachers is for school staff only. **Claude:** Ages 18 and over. In a school, for teachers only. Rules change often: check your school's rules and your national rules before you start. You never need AI to use a move.